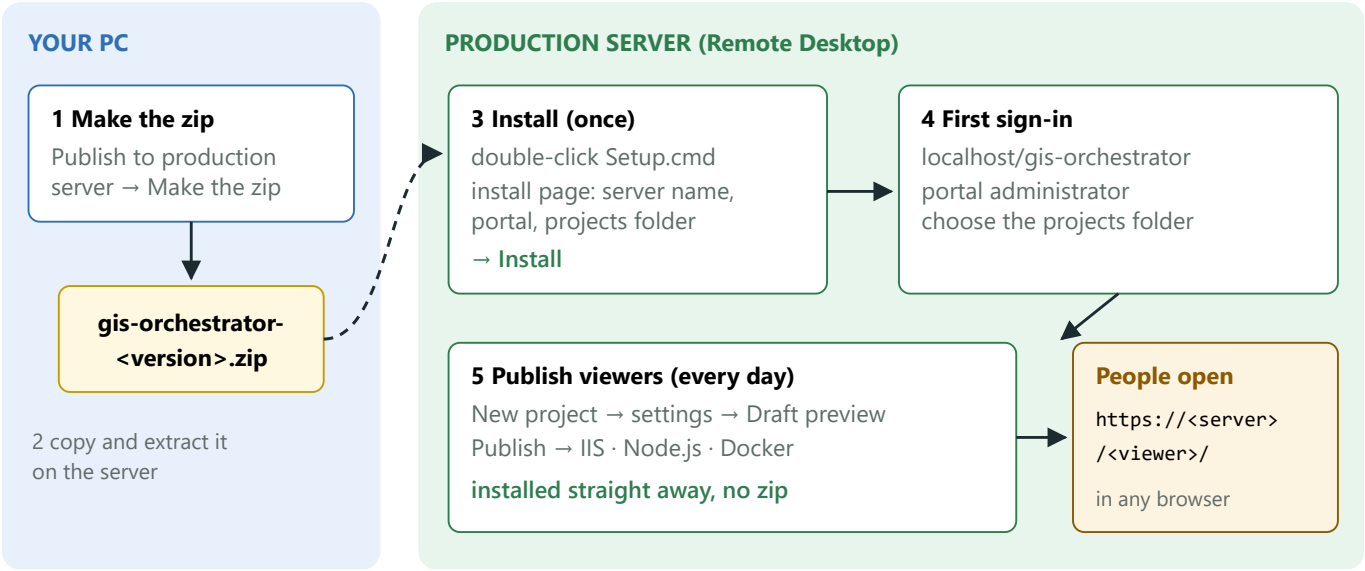
Orchestrator for ArcGIS Install Guide

Move the Orchestrator from your PC to the production server, install it, and publish your first viewer.



In this guide

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> **Two things to keep in mind.** The server name is typed **once**, on the install page; Publish never asks for it. And the only zip is the one you make on your PC: publishing a viewer writes it straight into IIS (or Node.js, or Docker) on the server.

1. Before you start

You need	Why
Windows Server (2016 or later; Windows 10/11 to try it)	The Orchestrator and its viewers run in IIS. Setup switches on the IIS features itself.
Administrator rights on the server	Setup installs into IIS, creates the projects folder and the install helper.
Remote Desktop to the server	You use the Orchestrator **on the server**, in its own browser (see the note below).
Edge or Chrome on the server	Saving projects to a folder needs a Chromium browser. Published viewers work in any browser.
Internet access from the server to `js.arcgis.com` and your portal	The map library loads from js.arcgis.com; web maps and sign-in come from your portal.
A **portal administrator** account (ArcGIS Online or Enterprise)	Only your portal's administrators may sign in to the Orchestrator and publish.
The server's name, e.g. `gis.example.gov.za`	The address people type to open viewers. Ask IT for the name in DNS.
Optional: an **https certificate** for that name	So viewers open over https (section 7). Without it they open over http.
Optional: **Node.js LTS** or **Docker** (Windows containers)	Only if you want viewers to run there instead of IIS.

> **Why on the server, at localhost?** The Orchestrator saves projects with the browser's folder access, which works only on `http://localhost` or https, and shows the drives of the computer the browser runs on. Opened from your own PC it would save to *your* drives, and it couldn't reach the install helper. So open it over Remote Desktop, on the server, at `http://localhost/gis-orchestrator/`.

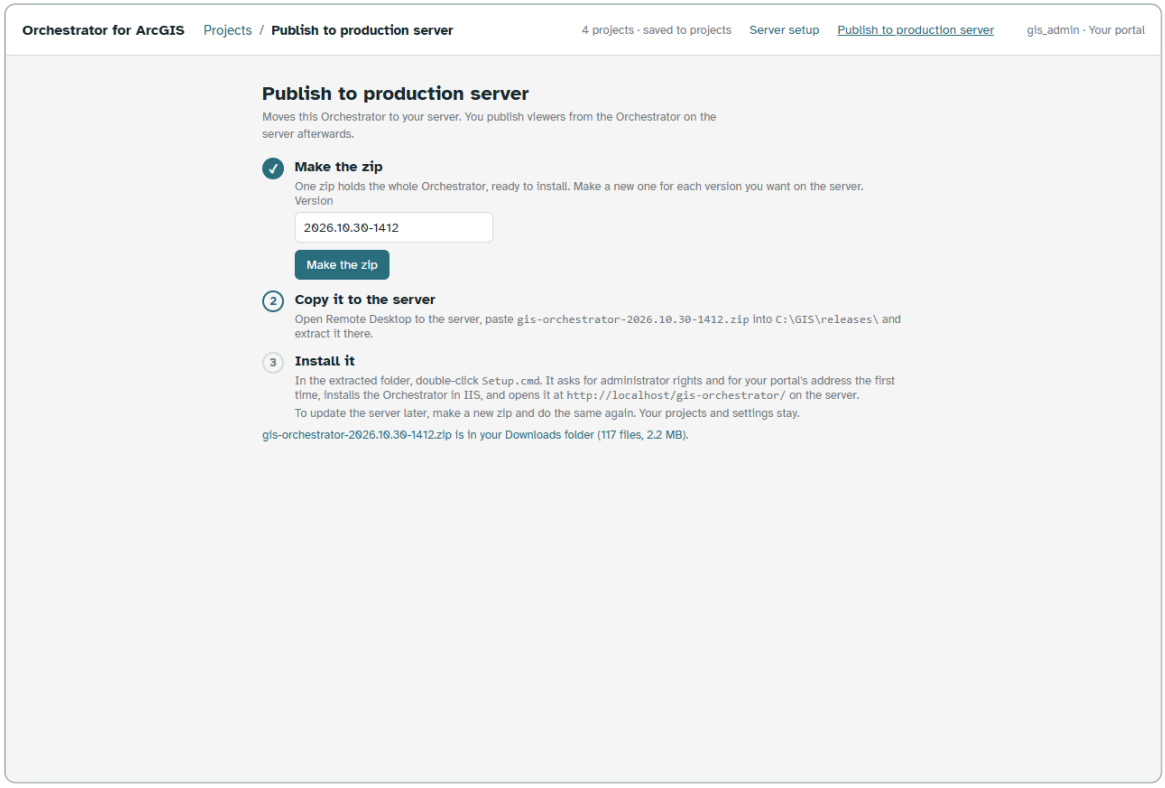
What gets installed where

Path	What
`C:\inetpub\gis-viewers\gis-orchestrator\v<version>\`	The Orchestrator, served at `/gis-orchestrator` under Default Web Site (one folder per version).
`C:\GIS\projects\` (your choice)	Projects, and every viewer you publish (`<project>\published\v<n>-<target>\`).
`C:\inetpub\gis-viewers\<viewer>\v<n>\`	Each published IIS viewer, one folder per version (so you can roll back).
`C:\ProgramData\GIS Orchestrator\`	Your answers (`authoring.json`, `server.json`) and the install helper. Administrators only.

YOUR PC

2. Make the zip

1. Start the Orchestrator on your PC: in the repository folder run `python tools/serve.py`, then open `http://localhost:8080/` and sign in.
2. Click **Publish to production server** in the header (next to Server setup).
3. Keep the version (today's date and time) or type your own, then click **Make the zip**.



Publish to production server: the zip is in your Downloads folder.

`gis-orchestrator-<version>.zip` holds the whole Orchestrator with `Setup.cmd` at its top. Make one for each version you want on the server. (From PowerShell, `.\tools\release.ps1` makes the same release.)

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3. Copy it to the server

1. Open Remote Desktop to the server.
2. Paste the zip into `C:\GIS\releases\` (any folder will do).
3. Right-click it → **Extract All...**. You get `C:\GIS\releases\gis-orchestrator-<version>\` with `Setup.cmd` inside.

> If Windows marks the files as downloaded, right-click the zip → **Properties** → tick **Unblock** before extracting.

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4. Install with the install page

1. In the extracted folder, double-click **Setup.cmd** and allow it administrator rights. A console window opens and stays open, and the **install page** opens in the browser. (If it doesn't, the console shows its address: copy it into Edge.)
2. Answer the four questions:

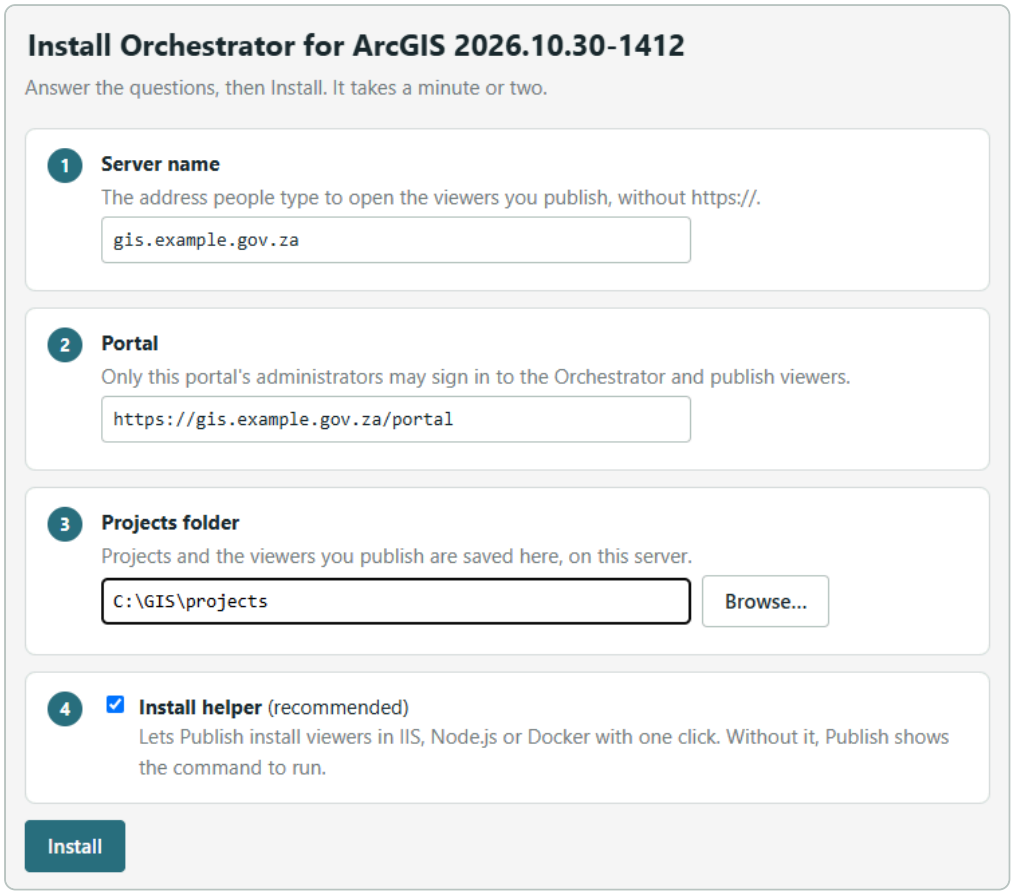
Question	What to type
Server name	The address people type to open viewers, without https:// or a path: `gis.example.gov.za`. It's filled in with the server's own name; change it if people use another.
Portal	Your portal: `https://gis.example.gov.za/portal` (ArcGIS Enterprise) or `https://<org>.maps.arcgis.com` (ArcGIS Online). Only its administrators may use the Orchestrator. If the server can't read the portal, the page asks for its **organisation id**: open `<portal>/sharing/rest/portals/self?f=json` in a browser and copy `"id"`.
Projects folder	`C:\GIS\projects`, or **Browse...** for another (the Windows folder picker may open behind the browser). Not under `C:\inetpub`, Windows or Program Files.
Install helper	Leave it ticked. It lets Publish install viewers with one click. Without it, Publish shows the command to run.

3. Click **Install**. Each step's output shows as it runs (a minute or two; longer the first time, while Windows adds IIS).
4. When it says **Done**, click **Open Orchestrator for ArcGIS**. The console window says you can close it.

A wrong answer is marked on its own field. The install's log is also kept in `%TEMP%\gis-orchestrator-setup-<date>.log`. If the log warns that Default Web Site has **no https binding** for the name, viewers still work, over http; section 7 adds https.

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4. Install with the install page (continued)



The install page: server name, portal, projects folder, install helper.

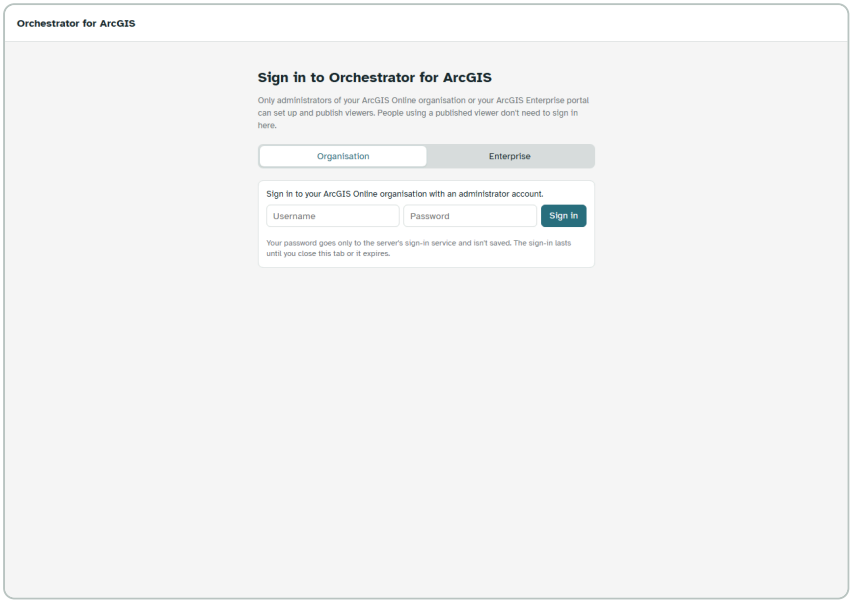


Done: each step's output, then Open Orchestrator for ArcGIS.

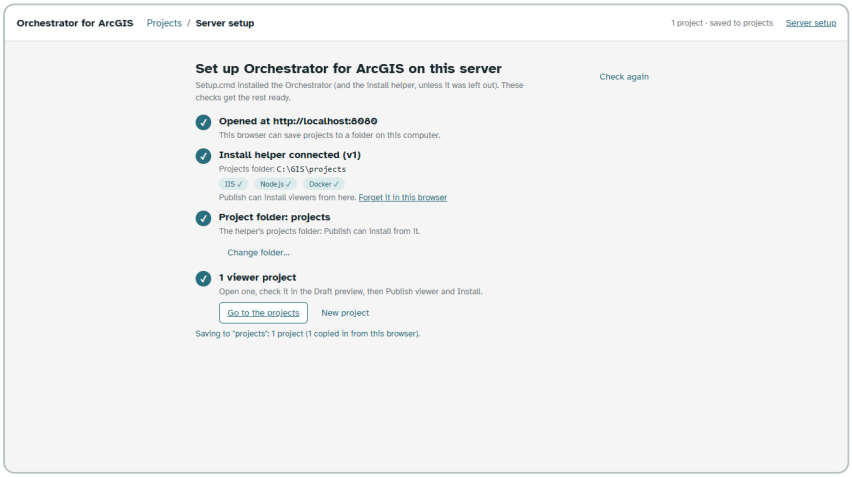
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5. First sign-in and Server setup

1. The Orchestrator opens at `http://localhost/gis-orchestrator/`. Sign in with your **portal administrator** account (or **Sign in with my Windows account** on a portal that uses Windows sign-in).
2. The **Server setup** checklist opens. Click **Choose project folder…** and pick the projects folder from the install page (`C:\GIS\projects`). The browser asks for permission once: allow it.
3. Every line should show a tick: opened on this server, install helper connected, project folder chosen.



Only your portal's administrators get in.



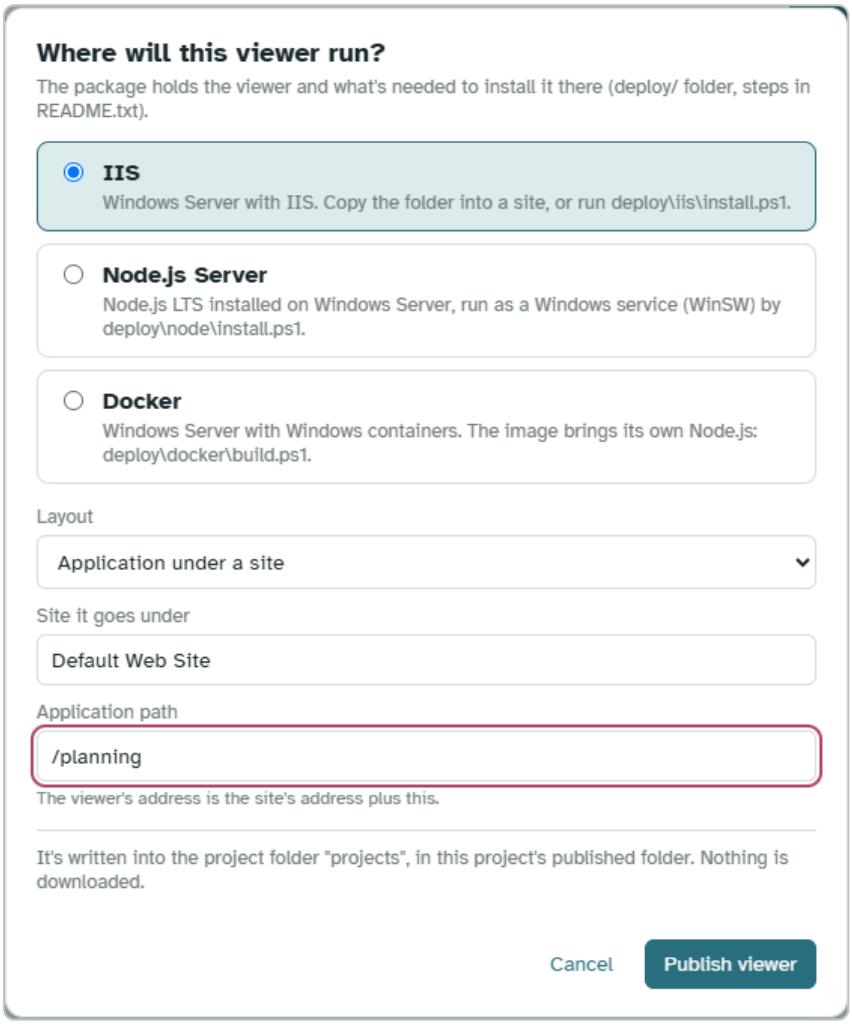
The Server setup checklist (header link **Server setup** opens it again).

> Browsers forget the project folder when they restart. If Publish later says **Choose the project folder first**, pick the same folder again.

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6. Publish a viewer

1. **New project** → paste a web map's item id or link → **Load web map**. Set it up from the settings rail.
2. **Update preview** → **Draft preview** to check it on the map.
3. Click **Publish viewer** and choose where it runs. **IIS** is selected: Default Web Site at `/<viewer>`. Click **Publish viewer**.
4. It's written into the projects folder and **installed straight away**. The dialog shows the log, then **Live at** the address people use, with **Copy**.



Where will this viewer run? IIS is selected.

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6. Publish a viewer (continued)



Installed straight away: Live at https://<server>/<viewer>/ with Copy, and the localhost address for the server itself.

Runs on	Asks for	People open it at
IIS	nothing (or another site or path)	`https://<server>/<viewer>/` (http until the site has a certificate)
Node.js	service name, port	`http://<server>:<port>/`. Needs Node.js LTS and WinSW on the server.
Docker	image name, port, Windows version	`http://<server>:<port>/`. Needs Docker with Windows containers.

Publish again whenever you change the viewer: each publish is a new version, and the dialog offers **Roll back to the version before**. For Node.js or Docker on a server without them, the dialog warns and waits for you to click **Install**.

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7. https for viewers

Setup checks whether Default Web Site has an https binding on port 443 for the server name. Without one, the **Live at** links are http. To switch to https:

1. Import the server's certificate (from IT) into **Local Computer** → **Personal**.
2. IIS Manager → **Default Web Site** → **Bindings...** → **Add**: type **https**, port **443**, host name your server name, and the certificate.
3. Double-click **Setup.cmd** again and click **Install again** (or run `Setup.cmd -Server <name>`). Its log now says viewers open at https.

Viewers you already published keep working; only the links Publish shows change. Nothing about the server name is stored in a viewer.

8. Updating and rolling back

Update the Orchestrator

1. On your PC: **Publish to production server** → **Make the zip** (a new version).
2. On the server: extract it next to the old one and double-click its **Setup.cmd**.
3. The page starts from your earlier answers. Click **Update**, then **Open Orchestrator for ArcGIS**.

Projects, published viewers, your answers and the browser's pairing with the helper all stay. After an update, publish viewers again to pick up the new version.

Roll back

What	How
A viewer	Publish dialog → **Roll back to the version before**, or in an administrator PowerShell in its published folder: `.\deploy\iis\install.ps1 -Rollback`
The Orchestrator	In an administrator PowerShell in any release folder: `.\deploy\iis\install.ps1 -Rollback`

9. Troubleshooting

You see	Do this
The install page didn't open	Copy the address from Setup's console window (`http://127.0.0.1:<port>/#key=…`) into Edge. Keep the console open.
"This page is from an earlier run of Setup.cmd"	Close the tab and use the page Setup opened this time (each run has a new key).
"The install page stopped"	It stops after 30 minutes without use, or when its console closes. Double-click Setup.cmd again.
Portal: "Couldn't read the organisation id"	The server can't reach the portal's `portals/self`. Type the organisation id the page now asks for (section 4), or fix the server's access to the portal.
"The install stopped (exit code …)"	Read the log on the page (also in `%TEMP%`), fix the cause, click **Back to the answers** and Install again.
Sign-in refused	Only administrators of the portal from the install page get in. To change the portal: run Setup.cmd again and click **Change the portal**.
Server setup: "Install helper not connected"	Run `deploy\setup\pair.cmd` from the release folder (pairs this browser again), or check the task **GIS Orchestrator install helper** in Task Scheduler.
Publish: "Choose the project folder first"	The browser forgot it after a restart: choose `C:\GIS\projects` again.
Publish shows a command instead of installing	No helper, or the chosen folder isn't the helper's projects folder. Choose that folder, or run the command shown in an administrator PowerShell.
Live at shows http://	No https binding for the name: section 7.
Viewer opens but the map is blank	The PC opening it can't reach `js.arcgis.com` or the portal, or the web map isn't shared with them.

10. Setup.cmd options

Double-clicked, Setup.cmd opens the install page. With an option it runs in its console window instead, asking there for anything it needs.

Command (from the release folder)	Does
`Setup.cmd -Console`	Install or update in the console, without the page.
`Setup.cmd -Server gis.example.gov.za`	Change the server name (and check https again).
`Setup.cmd -AuthorPortal <address>[,<address>]`	Change whose administrators may author viewers (more than one portal is fine).
`Setup.cmd -ProjectsRoot D:\GIS\projects`	Use another projects folder.
`Setup.cmd -ProjectsWriters 'DOMAIN\GIS Publishers'`	Let a group publish into the projects folder too.
`Setup.cmd -NoHelper` / `-RemoveHelper`	Install without the helper / remove it (the Orchestrator stays).
`Setup.cmd -PairOnly` (or `pair.cmd`)	Open the Orchestrator paired with the helper again, e.g. in another browser.

> **Safety, in short.** The install page and the install helper listen on `127.0.0.1` only, so other computers can't reach them, and each answers only requests with its own key. No password, token or key is written into any project or viewer. The full rules are in `deploy\README.md` in the release folder.